Week 11: Recurrent Neural Networks

Deep Learning – IITM BS

Comprehensive Notes + Practice Problems

Contents

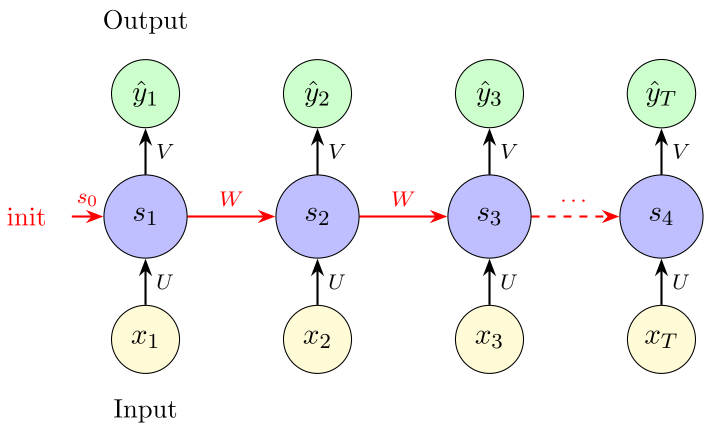
1 Why Sequential Models?

- Feedforward networks assume **fixed-size input** and **independence** between samples
- Many tasks have **variable-length** inputs: text, speech, time series
- Successive inputs are **dependent**: "The cat sat" → next word likely a preposition
- Need models that process sequences **step by step**, maintaining a **memory** (state)

Examples: POS tagging (output at each step), sentiment analysis (output at final step), machine translation (sequence to sequence).

2 Vanilla RNN

2.1 Architecture



RNN Equations

$$\boxed{s_t = \sigma(U \cdot x_t + W \cdot s_{t-1} + b)}$$

$$\boxed{\hat{y}_t = O(V \cdot s_t + c)}$$

$x_t \in \mathbb{R}^n$	input at time t	$U \in \mathbb{R}^{d\times n}$ (input → hidden)
$s_t \in \mathbb{R}^d$	hidden state at time t	$W \in \mathbb{R}^{d\times d}$ (hidden → hidden)
$\hat{y}_t \in \mathbb{R}^k$	output at time t	$V \in \mathbb{R}^{k\times d}$ (hidden → output)
$b \in \mathbb{R}^d$	hidden bias	$c \in \mathbb{R}^k$ (output bias)

Note: σ is typically sigmoid or tanh. O is the output activation (softmax for classification, linear for regression).

RNN Parameter Count

$$\text{Total} = \underbrace{n \cdot d}_{U} + \underbrace{d \cdot d}_{W} + \underbrace{d \cdot k}_{V} + \underbrace{d}_{b} + \underbrace{k}_{c}$$

KEY: Parameters are **shared across all time steps**!
Sequence length T does **NOT** affect parameter count.

2.2 Forward Pass: Step by Step

RNN Forward Pass Example

$U = 1$, $W = 1$, $s_0 = 0$, activation = sigmoid. Inputs: $x_1 = -\ln 3$, $x_2 = -0.25$.

Step 1: $s_1 = \sigma(U \cdot x_1 + W \cdot s_0) = \sigma(1 \cdot (-\ln 3) + 1 \cdot 0) = \sigma(-\ln 3)$
$= \frac{1}{1+e^{\ln 3}} = \frac{1}{1+3} = \mathbf{0.25}$

Step 2: $s_2 = \sigma(U \cdot x_2 + W \cdot s_1) = \sigma(-0.25 + 0.25) = \sigma(0) = \mathbf{0.5}$

Output: $V = 2$, linear output: $\hat{y} = V \cdot s_2 = 2 \times 0.5 = \mathbf{1.0}$

2.3 Backpropagation Through Time (BPTT)

- Total loss $= \sum_{t=1}^{T} L_t$ (sum over all time steps)
- Gradient of V: standard backprop (no temporal dependency)
- Gradient of U: standard backprop at each time step
- Gradient of W: involves **chain of products through time**:

$$\frac{\partial L}{\partial W} = \sum_t \frac{\partial L_t}{\partial s_t} \prod_\tau \frac{\partial s_\tau}{\partial s_{\tau-1}} \frac{\partial s_1}{\partial W}$$

- The product $\prod \frac{\partial s_\tau}{\partial s_{\tau-1}}$ causes **vanishing** (if < 1) or **exploding** (if > 1) gradients
- **Truncated BPTT**: limit backprop to τ steps

The Core Problem

After t steps, information from time step $t - k$ gets so "morphed" that:

1. The original information is hard to recover (forward problem)
2. The gradient signal can't assign credit properly (backward problem)

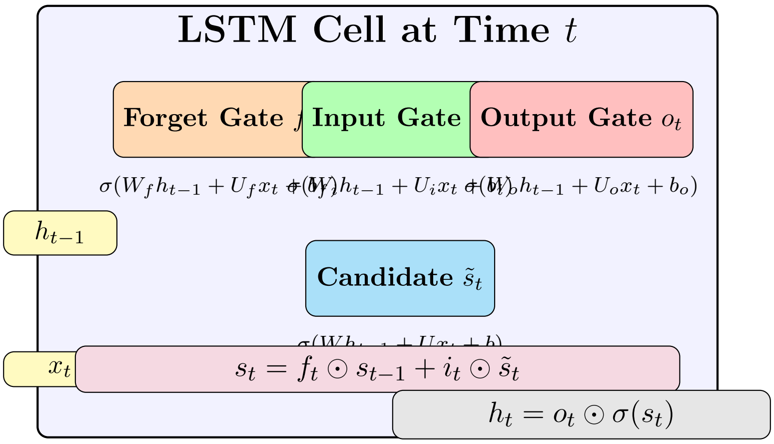
This motivates LSTM and GRU.

3 LSTM (Long Short-Term Memory)

3.1 Motivation

- We want to **selectively forget** irrelevant info (stop words)
- We want to **selectively read** important past info
- We want to **selectively write** new info to the state
- Solution: introduce **gates** that control information flow

3.2 LSTM Cell Architecture



LSTM Equations – MUST MEMORIZE

Three gates (each with sigmoid activation → values in $[0, 1]$):

$$f_t = \sigma(W_f\, h_{t-1} + U_f\, x_t + b_f) \qquad \text{(forget gate: what to discard)} \tag{1}$$

$$i_t = \sigma(W_i\, h_{t-1} + U_i\, x_t + b_i) \qquad \text{(input gate: what new info to store)} \tag{2}$$

$$o_t = \sigma(W_o\, h_{t-1} + U_o\, x_t + b_o) \qquad \text{(output gate: what to output)} \tag{3}$$

State update:

$$\tilde{s}_t = \sigma(W\, h_{t-1} + U\, x_t + b) \qquad \text{(candidate new state)} \tag{4}$$

$$s_t = f_t \odot s_{t-1} + i_t \odot \tilde{s}_t \qquad \text{(cell state: selective forget + read)} \tag{5}$$

$$h_t = o_t \odot \sigma(s_t) \qquad \text{(hidden output: selective write)} \tag{6}$$

$\odot$ denotes element-wise (Hadamard) product.

3.3 LSTM Parameter Counting

LSTM Parameters – EXAM CRITICAL

4 equations (3 gates + 1 candidate), each has:

$W \in \mathbb{R}^{d\times d}$	hidden-to-hidden weights: d^2
$U \in \mathbb{R}^{d\times n}$	input-to-hidden weights: nd
$b \in \mathbb{R}^{d}$	bias: d

Per gate: $nd + d^2 + d$. Four gates/equations: $4(nd + d^2 + d) = 4d(n + d + 1)$
Output layer: $V \in \mathbb{R}^{k\times d}$, $c \in \mathbb{R}^k \to dk + k = k(d + 1)$

$$\boxed{\text{Total LSTM params} = 4d(n + d + 1) + k(d + 1)}$$

LSTM has $\sim$4$\times$ the parameters of vanilla RNN!
T (sequence length) does **NOT** affect the count.

LSTM Param Count Example

$x_t \in \mathbb{R}^3$, $h_t \in \mathbb{R}^4$, $\hat{y}_t \in \mathbb{R}^3$, $T = 5$.
$n = 3$, $d = 4$, $k = 3$.
$4d(n + d + 1) = 4 \cdot 4 \cdot (3 + 4 + 1) = 16 \cdot 8 = 128$
$k(d + 1) = 3 \cdot 5 = 15$
Total $= 128 + 15 =$ **143**
Note: $T = 5$ is irrelevant!

3.4 Why LSTM Solves Vanishing Gradients

- Cell state s_t has an **additive** update: $s_t = f_t \odot s_{t-1} + i_t \odot \tilde{s}_t$
- Unlike RNN where $s_t = \sigma(\dots)$ (multiplicative chain), the additive structure preserves gradients
- If forget gate $f_t \approx 1$, information flows unchanged through many steps
- If s_{t-1} didn't contribute to s_t (forget gate ≈ 0), gradient into s_{t-1} vanishes—but *this is fine* because s_{t-1} wasn't responsible

4 GRU (Gated Recurrent Unit)

GRU Equations

$$
\begin{aligned}
o_t &= \sigma(W_o\, s_{t-1} + U_o\, x_t + b_o) && \text{(reset gate)}\\
i_t &= \sigma(W_i\, s_{t-1} + U_i\, x_t + b_i) && \text{(update gate)}\\
\tilde{s}_t &= \sigma(W(o_t \odot s_{t-1}) + U\, x_t + b) && \text{(candidate)}\\
s_t &= (1 - i_t) \odot s_{t-1} + i_t \odot \tilde{s}_t && \text{(new state)}
\end{aligned}
$$

Key differences from LSTM:

- Only 2 gates (not 3) $\rightarrow$ fewer parameters
- No separate cell state—forget and input gates are *tied*: $(1 - i_t)$ and i_t
- Parameters: $3d(n + d + 1) + k(d + 1)$

	RNN	**LSTM**	**GRU**
Gates	0	3 (+ candidate)	2 (+ candidate)
Cell state	No	Yes (s_t separate from h_t)	No
Param factor	1×	4×	3×
Vanishing gradient	Severe	Solved	Solved

5 Practice Problems

Problem 1: RNN Forward Pass [3 marks]

RNN with $U = 2$, $W = -1$, $s_0 = 0.5$, sigmoid activation, no bias.
Inputs: $x_1 = 0$, $x_2 = \ln 3$. Compute s_1 and s_2.

Problem 2: RNN Forward Pass (Vector) [3 marks]

$U = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $W = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $s_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, tanh activation, $x_1 = 1$ (scalar).
Compute s_1.

Problem 3: RNN Parameter Count [2 marks]

RNN with $x_t \in \mathbb{R}^5$, $s_t \in \mathbb{R}^8$, $\hat{y}_t \in \mathbb{R}^3$, with bias, $T = 100$. Total parameters?

Problem 4: LSTM Parameter Count [2 marks]

LSTM with $x_t \in \mathbb{R}^5$, $h_t \in \mathbb{R}^8$, $\hat{y}_t \in \mathbb{R}^3$, with bias, $T = 100$. Total parameters?

Problem 5: LSTM vs RNN [2 marks, MCQ]

Statement 1: Vanilla RNN has more trainable parameters than LSTM (same dimensions).
Statement 2: LSTM has more parameters because of multiple gates.
A. Both true, S2 is correct reason for S1 B. S1 true, S2 false
C. S1 false, S2 true D. Both false

Problem 6: Full RNN Pipeline [3 marks]

$U = 1$, $W = 1$, $V = 3$, $s_0 = 0$, sigmoid hidden activation, linear output, no bias.
Inputs: $x_1 = 0$, $x_2 = 0$, $x_3 = 0$.
Compute s_1, s_2, s_3 and $\hat{y}_3$.

Problem 7: GRU Parameter Count [2 marks]

GRU with $n = 4$, $d = 6$, $k = 3$, with bias. Total parameters?

Problem 8: Vanishing Gradient Fixes [2 marks, MSQ]

Which methods help address the vanishing gradient problem in deep networks? Select all correct.
A. Replace sigmoid with ReLU B. Increase training data C. Skip connections (ResNet)
D. Use LSTM instead of RNN E. Xavier/He initialization

Problem 9: Sequence Length and Parameters [2 marks]

An LSTM processes sequences of length 50 during training and length 200 during testing. Does the number of parameters change? Explain in one sentence.

Problem 10: Multi-step RNN [3 marks]

RNN with $U = 1$, $W = 2$, $s_0 = 0$, sigmoid activation, no bias.
Inputs: $x_1 = -\ln 3$, $x_2 = 0$, $x_3 = 0$.
$V = 1$, linear output. Compute final output $\hat{y}_3 = V \cdot s_3$.
Hint: Use $\sigma(-\ln 3) = 0.25$, $\sigma(0.5) \approx 0.622$, $\sigma(1.244) \approx 0.776$.

6 Solutions

Problem 1: $s_1 = \sigma(2(0) + (-1)(0.5)) = \sigma(-0.5) = \frac{1}{1+e^{0.5}} \approx \frac{1}{1+1.649} = \frac{1}{2.649} \approx \mathbf{0.378}$.
$s_2 = \sigma(2\ln 3 + (-1)(0.378)) = \sigma(2.197 - 0.378) = \sigma(1.819) \approx \frac{1}{1+e^{-1.819}} \approx \mathbf{0.860}$.

Problem 2: $s_1 = \tanh(U \cdot x_1 + W \cdot s_0) = \tanh\left(\begin{bmatrix}1\\0\end{bmatrix}(1) + \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}\begin{bmatrix}0\\0\end{bmatrix}\right) = \tanh\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}0.762\\0\end{bmatrix}$.

Problem 3: $n = 5, d = 8, k = 3$. Params $= nd + d^2 + dk + d + k = 40 + 64 + 24 + 8 + 3 = \mathbf{139}$. ($T$ irrelevant.)

Problem 4: $4d(n + d + 1) + k(d + 1) = 4(8)(5 + 8 + 1) + 3(9) = 32(14) + 27 = 448 + 27 = \mathbf{475}$. ($T$ irrelevant.)

Problem 5: **C**. S1 is false (LSTM has MORE params, $\sim 4\times$). S2 is true.

Problem 6: $s_1 = \sigma(0 + 0) = 0.5$. $s_2 = \sigma(0 + 0.5) = \sigma(0.5) \approx 0.622$. $s_3 = \sigma(0 + 0.622) = \sigma(0.622) \approx 0.651$. $\hat{y}_3 = 3 \times 0.651 = \mathbf{1.953}$.

Problem 7: $3d(n + d + 1) + k(d + 1) = 3(6)(4 + 6 + 1) + 3(7) = 18(11) + 21 = 198 + 21 = \mathbf{219}$.

Problem 8: **A, C, D, E**. Increasing training data does NOT fix vanishing gradients.

Problem 9: **No.** Parameters are shared across all time steps, so sequence length does not affect parameter count.

Problem 10: $s_1 = \sigma(-\ln 3) = 0.25$. $s_2 = \sigma(0+2(0.25)) = \sigma(0.5) \approx 0.622$. $s_3 = \sigma(0+2(0.622)) = \sigma(1.244) \approx 0.776$. $\hat{y}_3 = 1 \times 0.776 = \mathbf{0.776}$.